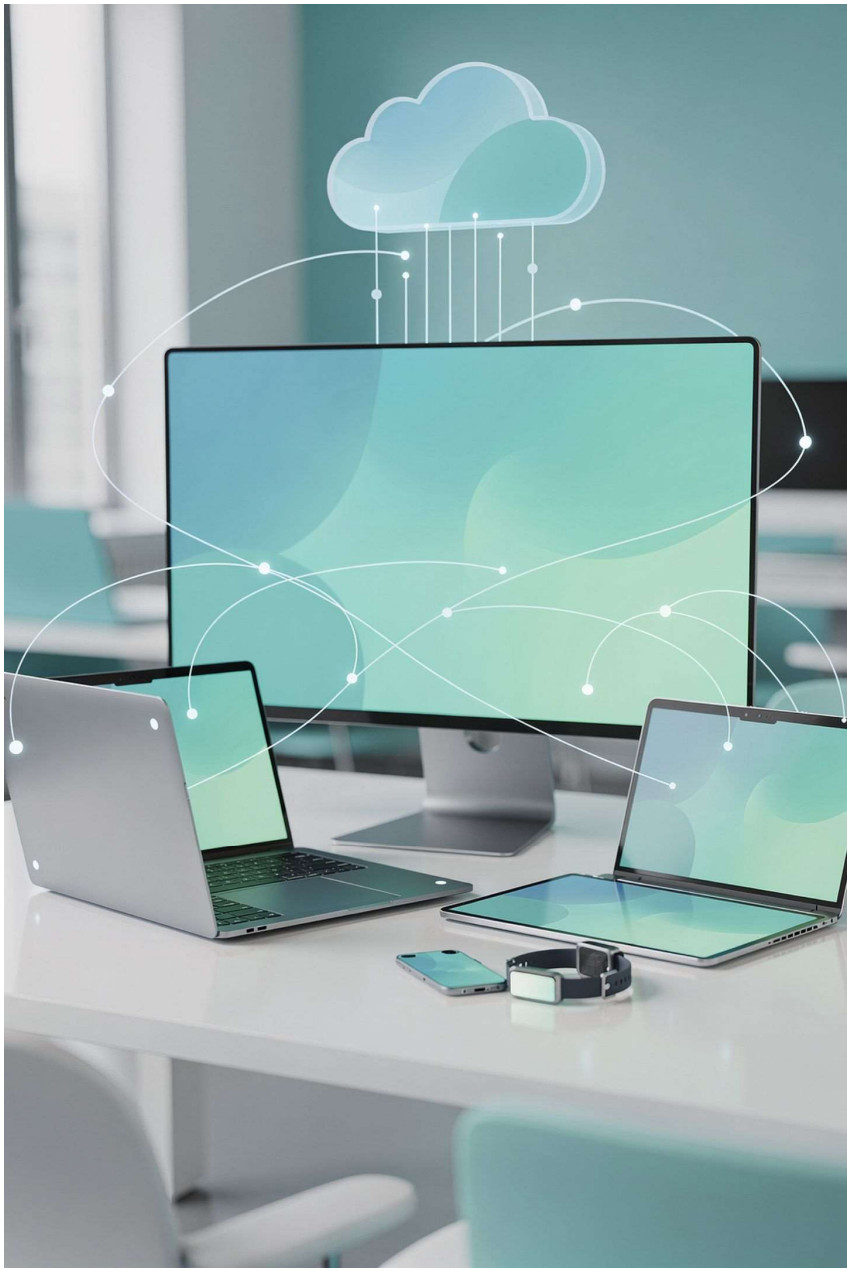
Introduction to Computer Systems

A foundational journey into how hardware, software, and people work together to process information. This presentation explores the building blocks of modern computing—from the physical components you can touch to the programs that bring them to life.

COMPUTER SCIENCE FUNDAMENTALS



FOUNDATIONAL CONCEPT

What Is Computing?

Computing (in Spanish/Basque, *Informática/Informatika —> Información Automática/Informazio Automatikoa*) refers to the automatic processing of information using electronic devices and computer systems.



It's important not to confuse an **Information System** with a **Computer System**. While related, they describe different things—one is the broader concept of organizing data, the other is the technology that processes it.

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System vs. System: A Key Distinction

Understanding the difference between these two terms is essential to building a solid foundation in information technology.

Information System

A set of elements aimed at processing and managing data and information—organized and ready for later use. Think of a filing cabinet system: structured, methodical, and built for retrieval.

Computer System

A set of parts or resources that interact with each other to store and process information toward a common goal. It is built from three pillars: hardware, software, and people.

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The Three Pillars of a Computer System

Every functioning computer system depends on the seamless interaction of three core elements working toward a shared purpose.



Hardware

The physical, tangible components—everything you can see and touch in a computer.



Software

The programs and data that tell the hardware what to do and how to do it.



People

The users and operators who interact with and direct the system toward its goals.

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WHY THEY MATTER

Objectives of a Computer System

Computer systems are designed with clear goals in mind—to make work faster, more reliable, and more efficient.

- **Reduce time, costs, and effort**

  Streamline operations across the entire system.

- **Modernize existing systems**

  Replace manual or outdated computerized processes.

- **Solve specific problems**

  Build new systems, possibly integrating them with existing ones.

- **Capture data at the source**

  Gather information directly where it originates.

- **Reduce manual tasks**

  Fewer manual steps means fewer possible errors.

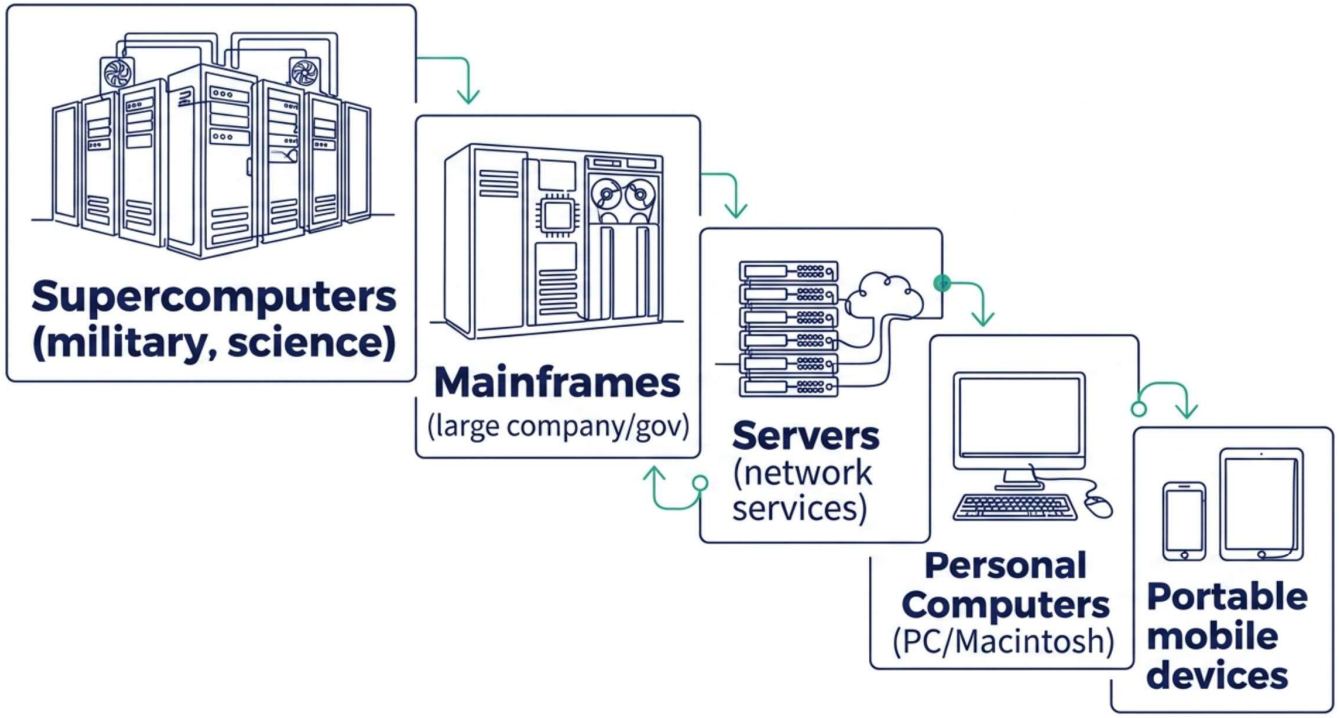
- **Centralize and boost productivity**

  Centralize process control to increase company output.

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Classification of Computer Systems

Not all computers are the same. They range from massive machines built for specialized tasks to the devices we carry in our pockets every day.



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A JOURNEY THROUGH TIME

Historical Evolution

Computers have transformed dramatically across four generations, each defined by a breakthrough in technology that made machines faster, smaller, and more affordable.

1st Generation (1940–1950)

1

Built for scientific and military use, such as the Mark I. Massive machines using vacuum tubes.

2nd Generation (1960–1965)

2

Transistors replaced tubes. Peripherals appeared, making computers faster and smaller.

3rd Generation (1966–1971)

3

Chips and integrated circuits arrived—faster, cheaper, and sold in greater numbers.

4th Generation (1971–1981)

4

The microprocessor was born, giving rise to the personal computer (PC). All basic elements on a single circuit.

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Hardware Components

Hardware refers to the electronic, mechanical, and electromechanical components of a computer system.

In short: hardware is **everything physical and tangible**—the parts you can hold in your hands.

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Input/Output Peripherals (I/O)

Peripherals are the bridge between the user and the operating system, handling the flow of information in both directions.

Input Peripherals

Send information **to** the operating system. Examples include the keyboard and mouse, which let users enter commands and data.

Output Peripherals

Receive information **from** the operating system. The monitor is a classic example, displaying results back to the user.



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Functional Elements

Inside every computer, a handful of key components work together to store, process, and move data.

CPU (Processor)

The brain that controls all operations.

RAM Memory

Temporary working memory for active tasks.

Input Peripherals

Devices that feed data into the system.

Hard Disk

Permanent internal storage for files.

External Storage

Removable units for extra capacity.

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HOW COMPONENTS COMMUNICATE

Buses: The Internal Highways

These functional elements are interconnected through **buses**—pathways that carry information between components. There are three main types.

Data Bus

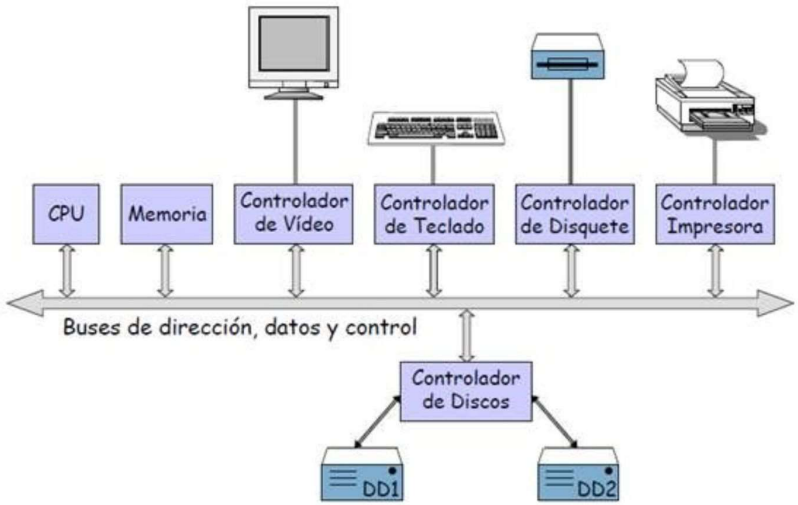
Carries the actual data being transferred between components.

Address Bus

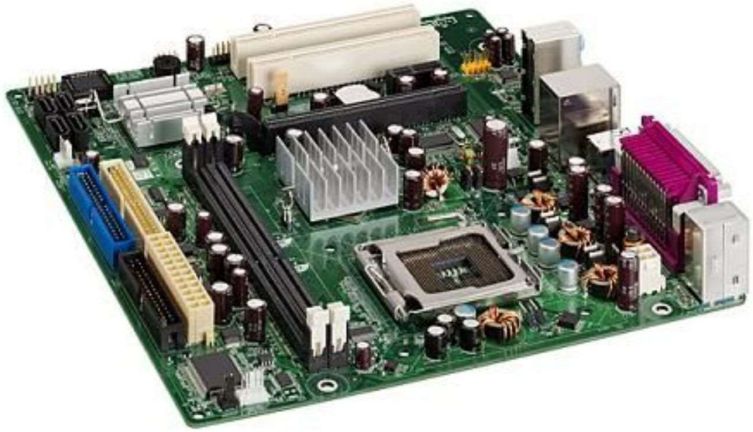
Carries information about **where** data should be sent or retrieved from.

Control Bus

Carries control signals that manage and coordinate all operations.



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The Motherboard

The motherboard is the central platform that connects all the major parts of the computer, allowing them to communicate as a unified system.

CPU

The processor socket sits at the heart of the board.

Main Memory

RAM modules plug directly into dedicated slots.

Input/Output Peripherals

Connectors link external devices to the system.

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THE BRAIN OF THE COMPUTER

Processor (CPU)

The CPU is the main component of the computer. It **controls all operations and tasks**, directing every other part of the system.

Control Unit (CU)

Interprets instructions and sends out the signals that coordinate the system.

Arithmetic Logic Unit (ALU)

Performs all arithmetic calculations and logical operations.

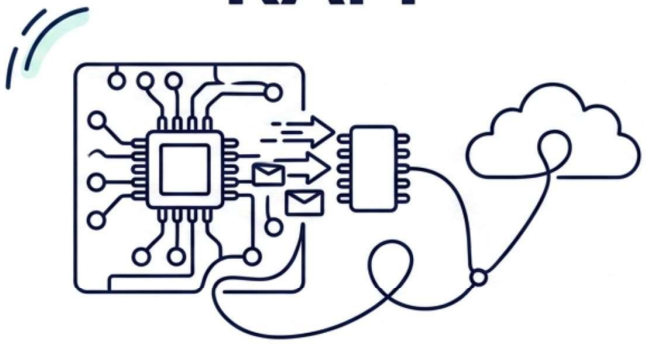


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Main Memory: RAM vs. ROM

A computer relies on two distinct types of main memory, each serving a very different purpose in keeping the system running.

RAM



Stores data to run programs.
Temporary & **Changeable**.
Works with storage devices.

ROM



Stores start-up info.
Non-Changeable.
Certain peripheral operations.

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THE INTANGIBLE SIDE

What Is Software?

Programs + Data

Software is a collection of programs and stored data that tells the computer what to do and how to do it.

Software performs the function of the program it implements. Without it, hardware is just inert metal and silicon.

- Works alongside external storage units
- Holds the instructions and data necessary to run a program
- Brings the physical machine to life with purpose

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Three Types of Software

Software is generally organized into three broad categories, each playing a unique role in the computing ecosystem.



System Software

Controls and manages the computer hardware and provides a platform for other software to run on.



Programming Software

Helps developers write, test, and debug the programs that power our world.



Application Software

Used by everyday users to perform specific, practical tasks.

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A Closer Look at Each Type

Let's explore what each software category includes, with a familiar real-world example for each.

System Software

Enables hardware to function.

- Operating systems
- Device drivers
- Diagnostic tools
- Optimization tools

Example: Microsoft Windows—manages hardware, files, and programs.

Programming Software

Used to develop programs.

- Text editors
- Compilers & interpreters
- IDEs (Integrated Development Environments)

Example: Visual Studio Code—write and edit code in many languages.

Application Software

Lets users perform specific tasks.

- Office & educational software
- Medical software
- Industrial control apps
- Numerical calculation tools

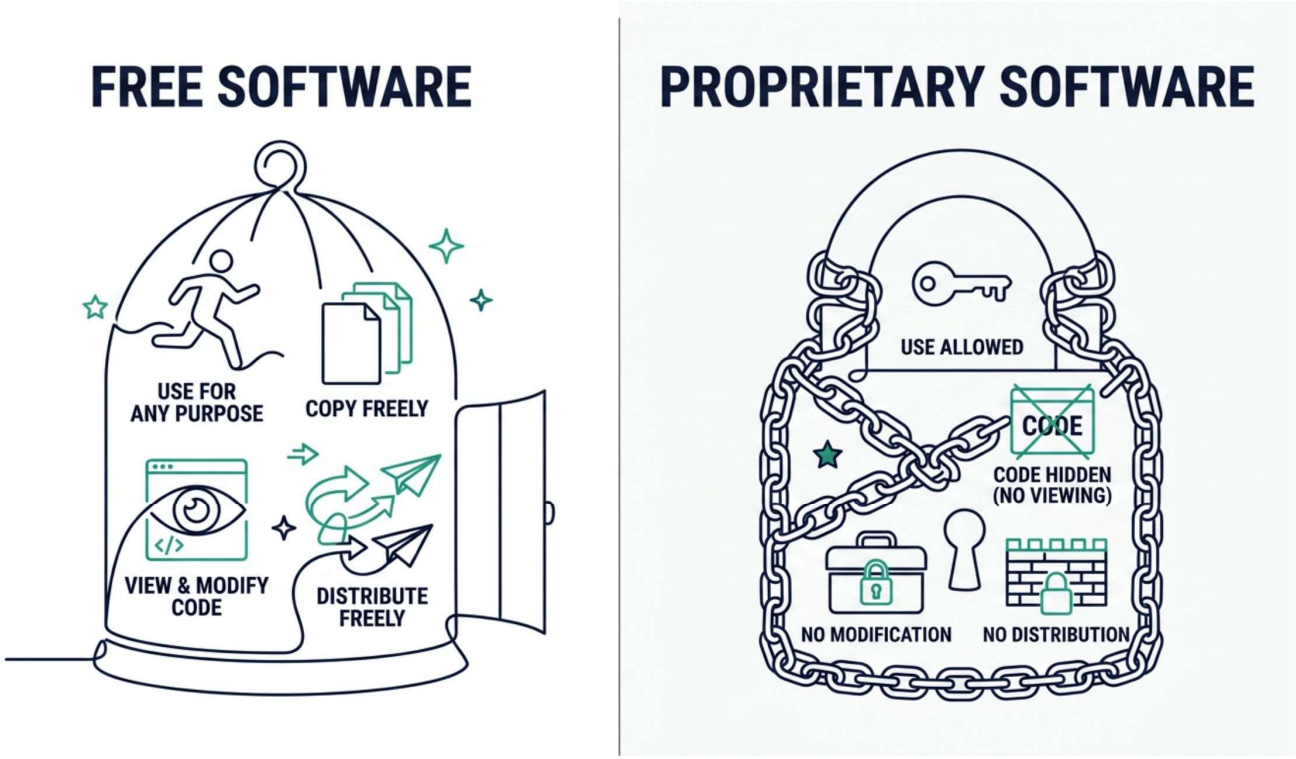
Example: LibreOffice—create and edit documents.

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FREEDOM VS. RESTRICTION

Free Software vs. Proprietary Software

Beyond its function, software is also defined by the rights it grants to users. This distinction shapes how software can be shared, studied, and improved.



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Software Licenses

Licenses are the legal agreements that govern how software may be used, copied, and shared. Four common types define the landscape.

1

Copyright

The work belongs to the author, who controls its use. Only personal use is allowed, and it cannot be distributed.

2

Copyleft

Can be distributed freely, ensuring all copies keep the same terms. Derived versions must also remain free.

3

Freeware

Free to distribute but not free software. Source code cannot be viewed or modified.

4

Shareware

Free for a trial period or with limited features, then requires payment to fully unlock and continue use.

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License Comparison at a Glance

This table summarizes the key rights and restrictions of each license type, making it easy to compare them side by side.

License	Distribute?	View Source?	Modify?	Cost
Copyright	No	No	No	Varies
Copyleft	Yes	Yes	Yes	Free
Freeware	Yes	No	No	Free
Shareware	Yes	No	No	Paid

Understanding these licenses helps you make informed, legal choices about the software you use, share, and build.

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